Nayunda Krisna
4522100071

No.
Date

1). $f(x) = \frac{(2a+3)x^3 - (b-a)x + c}{x^2+1}$

$a = 1$

$b = 5$

$c = 80$

→ $f(x) = \frac{(2(1)+3)x^3 - (5-1)x + 80}{x^2+1} = \frac{11x^3 - x + 80}{x^2+1}$

a). Hitung $f(1)$ & $f(-1)$

$f(1) = \frac{11(1)^3 - (1) + 80}{(1)^2+1} = \frac{11 - 1 + 80}{2} = \frac{90}{2} = 45$

$f(-1) = \frac{11(-1)^3 - (-1) + 80}{(-1)^2+1} = \frac{-11 + 1 + 80}{2} = \frac{70}{2} = 35$

b). google colab

c). Nilai a : mempengaruhi koefisien x^3, sehingga mengubah kemiringan grafik bag. kiri atau kanan.

Nilai b : mempengaruhi koefisien x sehingga menggeser bentuk fungsi lebih tajam atau lebih datar.

Nilai C : mempengaruhi nilai konstanta mengangkat grafik naik atau turun secara keseluruhan.

2). $\lim_{x \to a} = \frac{x^3 - (3a^2+b)x^2 + (3a^2b - c)x - a^3b + c}{x - a}$

$a = 4$

$b = 5$

$C = 80$

→ Hitungan bagian - bagian fungsi

- $3a^2 + b = 3(4)^2 + 5 = 3(16) + 5 = 48 + 5 = 53$
- $3a^2b - c = 3(16)(5) - 80 = 240 - 80 = 160$
- $a^3b - c = (64)(5) - 80 = 320 - 80 = 240$

→ bentuk limitnya :

$\lim_{x \to a} = \frac{x^3 - (3a^2+b)x^2 + (3a^2b - c)x - a^3b + c}{x - a}$

Scanned with CamScanner

→ menjadi :

$$\lim_{x \to 4} = \frac{x^3 - 53x^2 + 160x - 240}{x - 4}$$

→ turunan pembilang

$F'(x) = 3x^2 - 2(53)x + 160 = 3x^2 - 160x + 160$

→ substitusi $x = 4$

$F(4) = 3(16) - 106(4) + 160 = 48 - 424 + 160 = -216$

→ hasil limit

$$\lim_{x \to 4} = \frac{x^3 - 53x^2 + 160x - 240}{x - 4} = -216$$

→ $\lim_{x \to 0} = \frac{\sin(dx) - \tan(ex)}{x^3}$

- $d = 4$
- $e = 80$

→ pendekatan taylor

- $\sin(4x) = 4x - \frac{(4x)^3}{6} = 4x - \frac{64x^3}{6}$
- $\tan(80x) = 80x + \frac{(80x)^3}{3} = 80x + \frac{512000x^3}{3}$

→ disederhanakan

$$\lim_{x \to 0} = \sin(4x) - \tan(80x) = (4x - 80x) - \left(\frac{64x^3}{6} + \frac{512000x^3}{3}\right)$$

$$= -76x - \left(\frac{64}{6} + \frac{512000}{3}\right)x^3$$

→ Bentuk limitnya

$$\lim_{x \to 0} \frac{\sin(4x) - \tan(80x)}{x^3} = \frac{-76x - \left(\frac{64}{6} + \frac{512000}{3}\right)x^3}{x^3}$$

$$= \frac{-76x}{x^3} - \left(\frac{64}{6} + \frac{512000}{3}\right)$$

$$= \frac{-76}{x^2} - \text{konstanta}$$

$$= -\infty$$

⇒ Saat $x \to 0$, $\frac{-76}{x^2}$, maka hasilnya $-\infty$

3). $g(x) = \frac{\sin(ax) - ax\cos(bx)}{x^3}$

$a = 4$

$b = 5$

⇒ $g(x) = \frac{\sin(4x) - 4x\cos(5x)}{x^3}$

⇒ deret taylor

• $\sin(4x)$ sampai x^5

$\sin(4x) = (4x - \frac{(4x)^3}{3!} + \frac{(4x)^5}{5!} \cdot 4x - \frac{64x^3}{6} \quad \frac{1024x^5}{120}$

$= 4x - \frac{32x^3}{3} + \frac{128x^5}{15}$

$\cos(5x)$ sampai x^4

$4x\cos(5x) = 4x\left(1 - \frac{25x^2}{2} + \frac{625x^4}{24}\right) = 4x - 50x^3 + \frac{2500x^5}{24}$

⇒ Hitung pembilang

$g(x) = \sin(4x) - 4x\cos(5x) = \left(4x - \frac{32x^3}{3} + \frac{128x^5}{15}\right) - \left(4x - 50x^3 + \frac{2500x^5}{24}\right)$

→ dikelompokkan

$4x - 4x = 0$

$\frac{-32x^3}{3} + 50x^3 = \left(50 - \frac{32}{3}\right)x^3 = \frac{118}{3}x^3$

$\frac{128x^5}{15} - \frac{2500x^5}{24} = \left(\frac{128}{15} - \frac{2500}{24}\right)x^5$

→ $g(x) = \frac{\frac{118}{3}x^3 - \frac{22552}{240}x^5}{x^3} = \frac{118}{3} - \frac{22552}{240}x^2$

⇒ $\lim_{x \to 0} g(x) = \frac{118}{3} - \frac{22552}{240} \cdot 0 = \frac{118}{3}$

→ jika $\lim_{x \to 1} g(x) = \frac{118}{3}$

Scanned with CamScanner

No.

Date

4). 4+5+2+4+2+1+0+0+8+0 = 26

$h = \frac{26}{60} = 2,6$

$\Rightarrow \lim_{x \to 2,6} = \frac{\tan(x) - \tan(2,6)}{x - 2,6} = \sec^2(2,6) = 7.2401$

Scanned with CamScanner